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|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>1</b>  | Wages, salaries, and tips. This should be shown in box 1 of your Form(s) W-2. Attach your Form(s) W-2.                                                                                                                                                                                                                                                                                                               | <b>1</b>  |
| <b>2</b>  | Taxable interest. If the total is over \$1,500, you cannot use Form 1040EZ.                                                                                                                                                                                                                                                                                                                                          | <b>2</b>  |
| <b>3</b>  | Unemployment compensation and Alaska Permanent Fund dividends (see instructions).                                                                                                                                                                                                                                                                                                                                    | <b>3</b>  |
| <b>4</b>  | Add lines 1, 2, and 3. This is your <b>adjusted gross income</b> .                                                                                                                                                                                                                                                                                                                                                   | <b>4</b>  |
| <b>5</b>  | If someone can claim you (or your spouse if a joint return) as a dependent, check the applicable box(es) below and enter the amount from the worksheet on back.<br><input type="checkbox"/> <b>You</b> <input type="checkbox"/> <b>Spouse</b><br>If no one can claim you (or your spouse if a joint return), enter \$10,400 if <b>single</b> ; \$20,800 if <b>married filing jointly</b> . See back for explanation. | <b>5</b>  |
| <b>6</b>  | Subtract line 5 from line 4. If line 5 is larger than line 4, enter -0-.<br>This is your <b>taxable income</b> .                                                                                                                                                                                                                                                                                                     | <b>6</b>  |
| <b>7</b>  | Federal income tax withheld from Form(s) W-2 and 1099.                                                                                                                                                                                                                                                                                                                                                               | <b>7</b>  |
| <b>8a</b> | <b>Earned income credit (EIC)</b> (see instructions)                                                                                                                                                                                                                                                                                                                                                                 | <b>8a</b> |
| <b>b</b>  | Nontaxable combat pay election. <span style="float: right;">8b</span>                                                                                                                                                                                                                                                                                                                                                |           |
| <b>9</b>  | Add lines 7 and 8a. These are your <b>total payments and credits</b> .                                                                                                                                                                                                                                                                                                                                               | <b>9</b>  |
| <b>10</b> | <b>Tax.</b> Use the amount on <b>line 6 above</b> to find your tax in the tax table in the instructions. Then, enter the tax from the table on this line.                                                                                                                                                                                                                                                            | <b>10</b> |
| <b>11</b> | Health care: individual responsibility (see instructions)      Full-year coverage <input type="checkbox"/>                                                                                                                                                                                                                                                                                                           | <b>11</b> |
| <b>12</b> | Add lines 10 and 11. This is your <b>total tax</b> .                                                                                                                                                                                                                                                                                                                                                                 | <b>12</b> |

[Figure 2:](#) The instructions for filing taxes fit the definition of an algorithm.

How is it that taxes and parfaits can have anything in common? Taxes are inevitable, numeric, difficult, and universally disliked. Parfaits are infrequent, artistic, effortless, and adored without exception. The only trait they share is that people prepare both by following algorithms.

In addition to defining *algorithm*, the great computer scientist Donald Knuth noted that it is nearly synonymous with *recipe*, *procedure*, and *rigmarole*. In the case of filing taxes via the pictured 1040-EZ form, we have 12 steps (a finite list) that specify operations (like addition in step 4 and subtraction in step 6) to solve a specific type of problem: wanting to avoid being imprisoned for tax evasion. In the case of making a parfait, we have six finite steps that specify operations (like placing in step 1 and covering in step 2) to solve a specific type of problem: wanting to have a parfait in your hand or mouth.

As you learn more about algorithms, you will begin to see them everywhere and come to appreciate just how powerful they can be. In Chapter 1, we will discuss the remarkable human ability to catch a ball, and